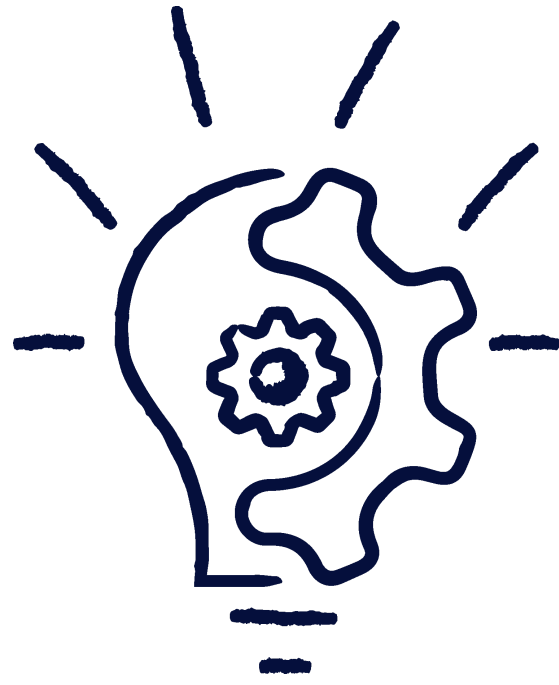
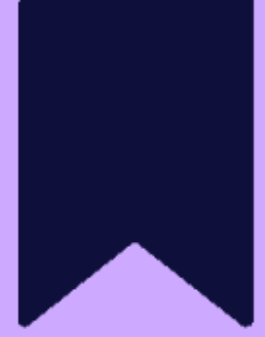




Course Code/Version: L-31/v1.0

Data Integrity and Data Governance



Study Guide

Table of Contents

[Study Guide](#)

[Introduction](#)

[Data Integrity Glossary](#)

[Who is responsible for data governance?](#)

[Paper-based, electronic, and hybrid systems](#)

[True copy](#)

[Raw data](#)

[Dynamic record vs static record](#)

[Data and metadata](#)

[Complete data](#)

[Audit trail](#)

[Electronic signature](#)

[Blank forms](#)

[ALCOA principles](#)

[Attributable](#)

[Legible](#)

[Contemporaneous](#)

[Original](#)

[Accurate](#)

[ALCOA+ principles](#)

[Complete](#)

[Consistent](#)

[Enduring](#)

[Available](#)

Introduction

Regulatory agencies have increasingly observed GxP violations involving data integrity and data governance during inspections in the Life Sciences industry.

This is cause for concern.

Ensuring data integrity is essential to preserving the safety, efficacy, and quality of drugs and medical devices—and to protecting public health.

GxP organizations must therefore take responsibility for the systems they use and the data they generate.

In this short course, I will explain some basic concepts involving data integrity.

We will discuss why data integrity is critical throughout the GxP data lifecycle, and how you can get full control of your data. You will learn the regulatory expectations for data integrity and data control. And finally, you will access a general overview of ALCOA and ALCOA+ terminology.

Data Integrity Glossary

Let's start with an overview of some important concepts.

The first concept is the life cycle of data.

What is the lifecycle of a document in GxP environments?

The **lifecycle of data** refers to how data is created, collected, stored, modified, processed, checked, used for decision-making, maintained, retrieved, transmitted, and ultimately discarded at the end of the retention period.

The second concept is data integrity.

Data integrity refers to the completeness, consistency, and accuracy of data, according to ALCOA principles (Attributable, Legible, Contemporaneous, Original, and Accurate).

We will discuss ALCOA and ALCOA+ in more detail in a later section.

Another definition of data integrity is the degree to which data is complete, consistent, accurate, trustworthy, and reliable—and the degree to which these characteristics are maintained throughout the data lifecycle.

Different definitions of data integrity share the same goal.

The objective of data integrity procedures is to have a system in place for the design and control that enables easy detection of errors, omissions, and aberrant results throughout the data lifecycle, either paper-based or electronic.

Now let's review a new term: data governance.

Data governance is the sum total of arrangements that provide assurance of data integrity.

Strong data governance ensures the right information, at the right level of quality, is available to the right person, for the right purpose, at the right time.

Basically, a data governance system is a special type of data management system that must be embedded into the GxP quality management system.

These arrangements should also ensure data, regardless of the process, format – for example: paper-based or electronic–, or technology used, complies with ALCOA+ principles throughout the data lifecycle.

This means you need to design and operate specific controls that are appropriate to the people, systems, and facilities involved. These controls must support a working environment that promotes data integrity.



For example, you need to guarantee that printed blank forms are unique and properly identifiable so that they can be reconciled upon completion.

In addition, you must guarantee **security access** for authorized employees through the use of a unique username and password.

Who is responsible for data governance?

Senior management is responsible for the implementation of systems and procedures that minimize the risk to data integrity, and for identifying the residual risk to data integrity using risk management principles.

Senior management must ensure all data used for batch release is complete, consistent, and accurate. Their responsibilities also include setting and maintaining the organization's quality culture, which is one of the pillars of having a Smart Quality management system.

Note that all employees of a GxP organization are involved in maintaining a good data governance system, and that people should be informed and trained accordingly.

Paper-based, electronic, and hybrid systems

In a previous lesson, we discussed how GxP companies can work on either paper-based systems, electronic systems, or a combination of the two, in what is known as a hybrid system.

All types of systems are acceptable, according to GxP standards, although it doesn't make sense to rely 100% on paper-based systems. Modern technology can ensure better data integrity and efficiency.

True copy

Regardless of the type of media used, a **true copy** is a copy of the original record that has been verified—with a dated signature or generation through a validated process—to have the same information as the original, including data that describes the context, content, and structure.

Copies of original paper records –for example: analytical summary reports, validation reports, etc.– are generally very useful for communication purposes—communication between companies operating at different locations, for example.

These records should be controlled during their lifecycle to ensure the data received from another site (e.g., a sister company, contractor, etc.) is maintained as a “true cop”.

Raw data

Raw data is the first-capture of information that is recorded on paper or electronically. It corresponds to the original data that has not yet been processed.

Raw data can be captured in a dynamic state or as a static record.

Dynamic record vs static record

A **static record** is a record format, such as a paper-based or electronic record, that is fixed and features little to no interaction between the user and the record content.



For example, a printed weighing balance record is a static record because the user cannot interact with or modify it. The same thing applies to a pH-meter printout. The printed data corresponds to the raw data itself.

A **dynamic record** is a record –for instance, an electronic record– that allows for an interactive relationship between the user and the record content.



For example, electronic records in database formats allow the user to track, trend, and query data. Chromatography records maintained as electronic records are another example that allow the user or reviewer—with appropriate access permissions—to change the baseline and reprocess the data so that the resulting peaks appear smaller or larger.



Another example of a dynamic record is a record that allows the user to modify formulas or entries in a spreadsheet that computes test results.

It is important to understand that when data is captured in a dynamic state and generated electronically, paper copies are not considered “raw data.”

Does this make sense? Great.

Now let’s review two additional terms: data and metadata.

Data and metadata

In previous courses, we discussed that data refers to all original records and true copies of original records (the source data and metadata).

Now it is time to review the meaning of metadata.

Metadata is the contextual information required to understand data.

Metadata is often described as “data about data.”

 Consider the following example: What is the meaning of the number 26?

26 by itself is meaningless without additional information. It might refer to 26 apples, 26 liters, or 26 mg.

Metadata is structured information that describes, explains, or otherwise makes it easier to retrieve, use, or manage data.

So in the above example, if 26 refers to “26 mg,” mg is the metadata. This allows us to understand the meaning behind 26.

 Another example of metadata are the date/time stamps we see. These document when the data was acquired. Another example is a user ID, which identifies the person performing a task. Audit trails are yet another example of metadata.

The relationship between data and metadata should be preserved in a secure and traceable manner throughout the retention period.

Complete data

Complete is one of the ALCOA+ criteria. Completeness is also an FDA GMP regulatory requirement in 21 CFR 211.194(a).

21 CFR 211.194: (a) Laboratory records shall include **complete data** derived from all tests necessary to assure compliance with established specifications and standards, including examinations and assays (...)

Put simply, complete data consists of every item of source data and metadata collected by a computerized system during an analysis, from sampling to generation of the reportable result.

Complete data includes any instrument log entries, documentation of mistakes, investigations, deviations, and instrument and calibration failures.

Audit trail

An **audit trail** is a secure, computer-generated, time-stamped electronic record that allows for the reconstruction of the course of events relating to the creation, modification, or deletion of an electronic record.

In previous lessons, we described an **audit trail** as an electronic record that is secure, computer-generated, and time-stamped, and that allows for the reconstruction of the course of events relating to the creation, modification, or deletion of an electronic record.

Electronic signature

An **electronic signature** is a computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual. It is the legally-binding equivalent of the individual's handwritten signature.

An electronic signature has the following characteristics:

- It is attributable to an individual.
- The act of "signing" is recorded in a computerized system, meaning it cannot be altered or manipulated.
- The record of the signature is associated with the entry made and how it can be verified.
- The electronic signature is secure, meaning it can only be applied by the "owner" of that signature.

Blank forms

Blank forms are working copies of documents for routine use in a GxP environment. They are copies obtained from a master document.

Examples of blank forms include:

- Electronic worksheets
- Analytical records
- Batch records
- Logbook forms
- Laboratory notebooks

Data integrity rules emphasize the importance of ensuring copies of GxP documents are issued and reconciled for use in a controlled and traceable manner.

This means that blank forms should be stamped for official use. The stamp must be made electronically by a computerized system, such as an eQMS, or manually by a document control group.

Operators must use blank forms and identify any data reported. Data is entered contemporaneously, calculations are reported, etc.

The form is reviewed upon completion for accuracy, authenticity, and completeness—by a different person than the operator.

Issued copies of a form must be reconciled to detect any discrepancies, errors, or evidence of inauthenticity.

ALCOA principles

Some key concepts of Good Documentation Practices (GdocPs) are summarized by the acronyms ALCOA or ALCOA+. These terms are used to evaluate data risk assessments.

ALCOA+ is a common acronym that refers to the five characteristics of good data: Attributable, Legible, Contemporaneous, Original and Accurate. The “+” puts additional emphasis on being Complete, Consistent, Enduring, and Available.

Attributable

Attributable refers to information that is generated or captured in the record that is uniquely identified as having been executed by the originator of the data –for example: a person or computer system.

This is extremely important in manufacturing processes and drug research, in GxP environments where every data point must be accounted for in order to proceed with further approval processes.

You should be able to identify who acquired the data or performed an action, as well as when the data originated and where the data came from –for example: a system, device, etc.

Keep in mind that having a person record something for someone else is only permitted under exceptional circumstances –in a sterile area, for example, or as an aid to foreign-language staff–. A supervisor is allowed to make the entries, but they must have witnessed the process taking place.

If a data point is altered, the person who performed it should be recorded, along with the time of action.

Legible

Legible refers to the data being readable, understandable, and traceable. It indicates a clear picture of the sequencing of steps or events in the record, with these elements recorded so that all conducted GxP activities can be fully reconstructed by people reviewing the records at any point during the record retention period.

Contemporaneous

Contemporaneous means the data is recorded at the time it is generated, captured, or observed. Back-dating and post-dating are not allowed. All captured data should have an associated date/time stamp.

Original

Original refers to the raw data that can be described as the first-capture of information, whether it is recorded on paper or electronically.

This data is used for further processing to fully reconstruct the conduct of the GxP activity. There should not be a secondary source of data.

💡 For example:

- Data should be directly recorded on a controlled blank paper form, and not on scratch papers or Post-Its.
- If a photocopy is made from an original document, the copy should be clearly distinguishable from the original document—for example, by using a stamp.

Accurate

Accurate means the data is correct, truthful, complete, valid, and reliable.

Data accuracy is the most critical part of the data quality framework. Data should be free from errors.

Any changes or updates to the data must be traceable and completed following Good Documentation Practices.

Redundant or duplicate data entries can create out-of-date records, resulting in poor data quality. Computerized systems should have redundant systems in place to verify the accuracy of the data.

In addition to ALCOA, there are several additional characteristics of good data (ALCOA+).

ALCOA+ principles

Complete

Complete means there is no deletion from the date of documenting.

Audit trails ensure all changes to the data are captured and indicate the source of the change as well as the time the change was made.

Nothing can be deleted or lost without a recorded reason. An audit trail ensures everything is captured as needed.

Consistent

Consistent means data points are time/date stamped, and that authorized readers can create a sequence of events based on the data.

Enduring

Enduring means there are systems where the data can be stored safely in reliable places for a long period of time, in readable condition, either as manual records or as a database.

Electronic data must be stored on controlled and robust electronic media—meaning CD-ROM, DVD-ROM, and uncontrolled USB media are not GxP-compliant or endurable.

Cloud-based solutions comply with the requirements.

Available

Available means the data must be available for review at any time during the retention period—for example, during the time of an audit or whenever it is required. Keep in mind that if you outsource GxP activities, data should remain available after the contract ends.

When the data does not reflect the above ALCOA+ principles during a risk assessment, specific remediation plans are required to fulfill the requirements.

That's all! Thank you for attending this course. I hope you found it useful and that you have learned what you expected. See you in other courses at Scilife Academy to continue your learning journey!